ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

A8ov5UqDdHWUAr8J81BCdUdoewNAPaGAtyH7TvQTYPoM

Date: 4/30/2025
Compounds: Amoxicillin, Caffeine, Aspirin

New Compound Analysis

Name: AmoSpirCaf
Formula: $C_{21}H_{35}N_5O_9$
Weight: 501.54
Drug-likeness: 0.75

Activity:
Antimicrobial, Analgesic, Stimulant

Target Analysis

Penicillin-binding proteins (PBPs)
Binding Affinity: 7.5
High relevance in bacterial infections

Cyclooxygenase (COX) enzymes
Binding Affinity: 8
Moderate to high relevance for pain and inflammation

Adenosine receptors
Binding Affinity: 6.5
High relevance for central nervous system stimulation

Toxicity Profile

Risk Level: Moderate
Affected Systems: Liver, Stomach, Heart

Mitigation Suggestions:
- Co-administration of proton pump inhibitors
- Monitoring of liver enzymes
- Cardiac function monitoring

Pharmacokinetics

Absorption:
Rapid oral absorption

Distribution:
Widely distributed, crosses blood-brain barrier

Metabolism:
Hepatic metabolism

Excretion:
Renal excretion

Half-life: 4-6 hours

Detailed Analysis

AmoSpirCaf is a hypothetical compound designed to combine the antimicrobial effects of amoxicillin, the analgesic properties of aspirin, and the CNS stimulating effects of caffeine. The compound features a complex molecular structure aimed at leveraging the beneficial effects of three pharmacological agents simultaneously. The mechanism of action involves the blockade of bacterial cell wall synthesis by targeting PBPs, inhibition of COX enzymes to reduce inflammation and pain, and antagonism of adenosine receptors to promote wakefulness and alertness. Although promising for conditions requiring multifaceted treatment approaches, AmoSpirCaf must be carefully evaluated for its pharmacokinetic and toxicity profile with special attention to potential gastrointestinal and cardiac effects. The drug is targeted for early-phase clinical trials contingent upon successful preclinical studies.
